

# Leveraging Blockchain Technology for Secure and Transparent Property Registry Management: Exploring Applications in Logistics Management, with Decentralized Decision-Making for Co-Ownership

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**Abstract**—Conventional property register systems often experience inefficiencies, lack transparency, and are prone to fraud. This study presents an innovative method using blockchain technology to create a secure, transparent, and efficient property register system. The system employs smart contracts on the Ethereum blockchain and integrates a Decentralized Autonomous Organization (DAO) framework for managing co-ownership. This enables property registration, co-ownership administration, proposal voting, and data retrieval. The contract complies with Ethereum Request for Comments 20 (ERC20) standards, allowing for fractional ownership via tokens. We examine the advantages of incorporating blockchain technology, DAO governance, and ERC20 functionality into property registries. Additionally, we explore potential applications in supply chain management, specifically in the logistics sector, to improve transparency and efficiency.

**Keywords**—Blockchain, Ethereum, Smart Contracts, ERC20, Property Registry, Logistics, DAO, Co-ownership

## I. INTRODUCTION

DAOs are blockchain-based applications for automating governance processes. They were first described as 'coded organizations' by Rikken et al. [1]. The code automates governance, eliminating the need for human intervention. This decentralization that decisions are executed without a trusted third party [9].

Blockchain, initially conceptualized for cryptocurrencies like Bitcoin [2], has evolved into a transformative technology with applications beyond digital currencies. Blockchain is a distributed database technology allowing instant, verified transactions without a central authority. It is a shared, immutable ledger that facilitates asset tracking in a business network [3]. Assets can be tangible or intangible and can be tracked and traded on a blockchain network, ensuring data integrity and security [4].

The real estate sector, known for complex transactions and reliance on intermediaries, can significantly benefit from

blockchain technology [6]. Traditional property register systems are often inefficient, inaccurate, and lack transparency, leading to conflicts and fraud. Blockchain offers a secure, transparent, and decentralized platform for recording property transactions and ownership information.

Tokenizing Real World Assets (RWAs), such as real estate, is possible with blockchain technology. Tokenization transforms tangible assets into digital tokens that can be exchanged on a blockchain platform [7]. This allows fractional ownership, enhancing liquidity and market participation. Tokenization democratizes access to real estate assets by dividing property ownership into smaller, tradable units [8]. Buterin described DAOs as a way to explore new forms of organizational governance integrated into a distributed ledger [9]. Dhillon et al. defined DAOs as blockchain organizations constructed through collective agreement among members [10]. Blockchain-based management reduces administrative costs by automating tasks [11]. DAOs facilitate interactions among a collective, where token holders participate like shareholders [14].

## II. CONTRIBUTION

We implemented a blockchain-based property registration system to enhance security, transparency, and efficiency in property management. The system uses Ethereum smart contracts for secure, immutable property ownership and transaction records. A DAO structure enables decentralized decision-making, reducing conflicts. ERC20 tokens allow fractional ownership, democratizing real estate investment. Our system can also improve logistics management by tokenizing warehouse ownership, facilitating investment, and enhancing property use and regulatory compliance. The DAO framework supports collaborative decision-making among investors [12]. This technology synergizes with supply chain management, improving tracking and land use monitoring, essential for regulatory compliance and operational efficiency. We showcase

our method's practicality with real-world scenarios in the real estate and supply chain sectors, demonstrating significant improvements in property record management. Our system offers a comprehensive framework to transform property registration using blockchain, DAOs, and ERC20 standards.

### III. TECHNOLOGIES AND STANDARDS USED

#### A. Blockchain and Ethereum

Blockchain is a distributed database technology enabling instant, verified transactions without a central authority. Transactions are stored as a distributed ledger among networked computers, ensuring continuous verification and data integrity [3]. Blockchain facilitates tracking and trading assets through smart contracts on Ethereum [4]. Smart contracts, written in Solidity, are automated, transparent, and immutable programs managing contract terms. They ensure data consistency and security across the network. Ethereum was chosen for its robust smart contract functionality, active developer community, and extensive ecosystem, making it ideal for a sophisticated property registry system [24].

#### B. ERC20 standard

ERC20 is a standard for creating fungible tokens on the Ethereum network. Fungible tokens are interchangeable and have a fixed value. ERC20 establishes uniform guidelines for token compatibility across platforms. Fractional ownership using ERC20 tokens increases accessibility and liquidity in the real estate market by dividing properties into smaller, tradable units [7]. This democratizes real estate investment, making it more accessible to a broader range of investors [8].

#### C. DAO

A DAO is an entity represented by rules encoded as a computer program, transparent, controlled by members, and not influenced by a central government. DAOs operate on blockchain technology and use smart contracts to enforce rules and execute decisions without intermediaries. In real estate, a DAO can manage co-ownership, enabling decentralized, democratic decision-making, enhancing accountability, and reducing conflicts among co-owners [9].

### IV. SYSTEM ARCHITECTURE

Our blockchain-based property registry system includes several critical components to ensure secure, transparent, and efficient property ownership and transactions. It uses Ethereum smart contracts to automate and enforce property management policies. All records are securely stored on the Ethereum blockchain, ensuring data transparency and immutability.

The `FractionalProperty` smart contract manages property registration, fractional ownership, and governance. Key aspects include co-ownership administration and ERC20 token integration. Property owners can register new properties, providing details like location, size, and total ERC20 tokens for fractional ownership. Co-ownership administration maintains a list of co-owners and allows adding new ones. ERC20 tokens facilitate fractional ownership and trading [5].

The system uses a DAO structure for decentralized decision-making among co-owners. The DAO includes a proposal and voting mechanism. Co-owners can create proposals related to property management, vote using their ERC20 tokens, and execute approved proposals automatically via smart contracts, ensuring transparency and reducing conflicts [13].

Development is done using Remix IDE, enabling direct interaction with smart contracts. Users can register properties, manage co-ownership, create and vote on proposals, and handle token transactions via Remix IDE. Off-chain storage solutions like (InterPlanetary File System)IPFS store large property metadata, ensuring accessibility without overloading the blockchain. The system can integrate with external property management systems and government registries for data accuracy and legal compliance [16].

This architecture ensures a robust, transparent property registry management solution leveraging blockchain, DAOs, and ERC20 tokens, enhancing security, transparency, and inclusivity in real estate investments [26].

### V. IMPLEMENTATION

The implementation involves critical components: property registration, co-ownership management, proposals and voting, and token operations. Each uses Ethereum smart contracts for secure, transparent, automated operations.

#### A. Property Registration

The `registerProperty` function in the `FractionalProperty` smart contract facilitates property registration. When a new property is registered, the following steps are executed:

**Incrementing Property Count:** The total number of properties (`totalProperties`) is incremented by one.

**Initializing Property Details:** A new `Property` struct is created, containing the property's location, size, characteristics, and the total number of ERC20 tokens representing fractional ownership.

**Assigning Ownership:** The property is initially assigned to the system owner (`msg.sender`), who is added to the list of co-owners.

**Minting Tokens:** ERC20 tokens corresponding to the property's fractional ownership are minted and assigned to the owner.

**Storing Property Information:** The new property details are stored in the `properties` mapping using the property ID.

**Event Emission:** An event (`PropertyRegistered`) is emitted to notify the system of the new property registration.

This process ensures all property details are securely recorded on the blockchain, providing an immutable, transparent record of ownership [20]. see algorithm 1.

#### B. Co-Ownership Management

The `addCoOwner` function manages co-ownership, allowing existing co-owners or the system owner to add new co-owners. Steps involved:

**Algorithm 1** Property Registration**Require:** location, size, characteristics, totalTokens**Ensure:** Property is registered and ERC20 tokens are minted

```

if totalTokens > 0 then
  Increment totalProperties
  Create a new Property with given details
  Assign msg.sender as the initial owner
  Mint totalTokens ERC20 tokens to msg.sender
  Store the new Property in properties mapping
  Add property ID to userProperties of msg.sender
  Emit PropertyRegistered event
else
  return error 'Total tokens must be greater than zero'
end if

```

**Verification:** Verifies the provided property ID is valid and the new co-owner is not already listed.

**Updating Co-Owners:** Adds the new co-owner to the list for the specified property.

**Updating User Properties:** Maps the new co-owner's address to the property ID in the userProperties mapping.

**Event Emission:** Emits an event (CoOwnerAdded) to notify the system of the new co-owner.

This function ensures the co-owners list is accurately maintained and changes are transparently recorded on the blockchain [17]. see algorithm 2.

**Algorithm 2** Add Co-Owner**Require:** id (uint), newCoOwner (address)**Ensure:** Boolean indicating success or failure

```

1: if id > 0 and id ≤ totalProperties then
2:   prop ← properties[id]
3:   if not isCoOwner(prop.coOwners, newCoOwner) then
4:     prop.coOwners.push(newCoOwner)
5:     userProperties[newCoOwner].push(id)
6:     Emit CoOwnerAdded(id, newCoOwner)
7:     return true
8:   else
9:     return error 'Address already a co-owner'
10:  end if
11: else
12:  return error 'Invalid property ID'
13: end if

```

*C. Proposals and Voting*

The DAO structure allows co-owners to create, vote on, and execute proposals related to property management:

**Creating Proposals:** The createProposal function allows any co-owner to create a proposal. The proposal includes a title, description, and voting duration. Once created, the proposal details are stored, and a ProposalCreated event is emitted.

**Voting on Proposals:** The voteOnProposal function enables co-owners to vote on proposals using their ERC20 tokens, where each token represents one vote. The function verifies the voter's token balance and ensures they have not voted previously. Votes are then tallied, and a Voted event is emitted.

**Executing Proposals:** After the voting deadline, the executeProposal function is called to automatically execute the proposal if it passes (i.e., receives the required number of votes). A ProposalExecuted event is emitted to indicate the execution of the proposal.

This decentralized and transparent voting mechanism ensures all co-owners have a voice in property-related decisions, enhancing accountability and reducing conflicts [23]. see algorithm 3;

**Algorithm 3** Create Proposal**Require:** title (string), description (string), votingDuration (uint)**Ensure:** Boolean indicating success or failure

```

1: Increment totalProposals
2: Initialize newProposal with given title, description,
   and votingDeadline (current time + votingDuration)
3: proposals[totalProposals] ← newProposal
4: Emit ProposalCreated(totalProposals, title,
   description, newProposal.votingDeadline)
5: return true

```

*D. Token Operations*

The system allows users to deposit and withdraw ERC20 tokens, representing fractional ownership of properties. The functions involved are:

**Deposit:** The deposit function allows users to deposit funds into the system, receiving an equivalent amount of ERC20 tokens. The tokens are minted and assigned to the user, and a Deposit event is emitted.

**Withdraw:** The withdraw function enables users to withdraw their funds by burning the equivalent amount of ERC20 tokens. The tokens are burned, and a Withdraw event is emitted to notify the system of the transaction.

These token operations facilitate the buying and selling of fractional property ownership, providing greater liquidity and accessibility in the real estate market [7]. see algorithm 4.

## VI. USE CASE SCENARIO

*A. Property Management in the Logistics Industry*

In the logistics industry, efficient property management is crucial for optimizing operations and reducing costs. By implementing a blockchain-based property registry system, logistics companies can enhance the management of their warehouses and other properties. Here's a concrete example of how this system can be used:

**Tokenization of Warehouses:** Logistics companies can tokenize their warehouses by creating ERC20 tokens that represent fractional ownership of each warehouse. For instance,

**Algorithm 4** Vote on Proposal**Require:** *proposalId* (uint), *vote* (bool)**Ensure:** Boolean indicating success or failure

```

1: if proposalId > 0 and proposalId ≤ totalProposals
   then
2:   proposal ← proposals[proposalId]
3:   if current_time < proposal.votingDeadline then
4:     if balanceOf(msg.sender) > 0 and not
       proposal.voted[msg.sender] then
5:       votingPower ← balanceOf(msg.sender)
6:       if vote then
7:         proposal.yeaVotes ← proposal.yeaVotes +
           votingPower
8:       else
9:         proposal.nayVotes ← proposal.nayVotes +
           votingPower
10:      end if
11:      proposal.voted[msg.sender] ← true
12:      Emit Voted(proposalId, msg.sender, vote)
13:      return true
14:    else
15:      return error 'You have already voted or have no
        voting power'
16:    end if
17:  else
18:    return error 'Voting deadline has passed'
19:  end if
20: else
21:   return error 'Invalid proposal ID'
22: end if

```

a large warehouse can be divided into 1,000 tokens, with each token representing a 0.1% ownership stake in the property. This allows multiple investors to own a share of the warehouse, democratizing access to real estate investments [21].

**DAO Governance for Decision-Making:** Using a Decentralized Autonomous Organization (DAO) structure, the co-owners of the warehouse can participate in the decision-making process. Key decisions such as maintenance, upgrades, and leasing can be managed through proposals and voting mechanisms. Here's how it works:

1) **Creating Proposals:** Any co-owner can propose improvements or changes to the warehouse by submitting a proposal through the DAO interface. For example, a proposal might suggest installing solar panels to reduce energy costs.

2) **Voting on Proposals:** Co-owners vote on the proposal using their ERC20 tokens, with each token representing one vote. This ensures that all stakeholders have a say in the decision-making process.

3) **Executing Proposals:** Once a proposal receives the required number of votes, it is automatically executed by the smart contract. For instance, if the solar panel installation proposal is approved, the smart contract initiates the process by allocating funds and hiring contractors.

## VII. BENEFITS AND CHALLENGES

## A. Benefits

Compared to conventional property management systems, the blockchain-based property registry system has a number of important advantages. Improved inclusivity, efficiency, transparency, and security are some of these advantages.

One of the main benefits of utilizing blockchain technology is increased security. Since data on blockchain is dispersed among several nodes due to its decentralized nature, it is extremely resistant to hacking and tampering. Once a property record is uploaded to the blockchain, cryptographic procedures guarantee that it cannot be removed or changed without the network's approval. Property ownership records are shielded from fraudulent activity and illegal alterations by this immutability [2].

Another important advantage of the method is transparency. Every transaction and change in property ownership is noted on a public ledger that is available to every network user. Since each party can independently confirm the accuracy of the data, this transparency lowers the possibility of disagreements and fraud. Because a property's whole transaction history is visible to all parties, it also fosters stakeholder trust [3].

The efficiency of property administration is greatly increased by the automation of numerous procedures through the use of smart contracts on the Ethereum blockchain. When specific criteria are satisfied, smart contracts automatically carry out predetermined operations, doing away with the need for middlemen and saving money and time on manual processing. Smart contracts facilitate the effective and timely handling of property transfers, co-ownership additions, and proposal voting, hence improving overall operational efficiency [4].

Integrity is one of the system's main advantages. Tokenizing assets with ERC20 tokens makes real estate investing more accessible to anyone. A greater variety of investors, including those with low cash, can engage in property ownership by dividing properties into smaller, tradable pieces. More people can profit from real estate investments thanks to this inclusivity, which makes investment opportunities more accessible to a wider audience and permits fractional ownership [7].

## B. Challenges

Despite the numerous benefits, the blockchain-based property registry system also faces several challenges, including scalability, regulatory compliance, and the need for thorough security audits.

Scalability is one of the primary challenges of blockchain technology. As the number of transactions increases, the blockchain can become congested, leading to slower transaction times and higher costs. Ethereum, in particular, has faced scalability issues with its current infrastructure. To overcome this, solutions such as layer-2 scaling (e.g., Plasma, state channels) and future upgrades like Ethereum 2.0 (which aims to introduce sharding) are being developed to improve transaction throughput and reduce latency [16].

Navigating the regulatory landscape is another significant challenge. Real estate transactions are subject to various local

and national laws, which can vary widely between jurisdictions. Ensuring compliance with these regulations while implementing a decentralized system requires careful consideration and potentially collaboration with regulatory bodies. Smart contracts need to be designed to comply with legal requirements, and there may be a need for hybrid systems that integrate both on-chain and off-chain components to satisfy regulatory demands [15].

While blockchain itself is secure, smart contracts can be vulnerable to bugs and exploits if not properly coded and audited. Thorough security audits are essential to identify and rectify vulnerabilities in the smart contracts before they are deployed. Additionally, ongoing monitoring and updates are necessary to address new security threats as they emerge. Employing best practices in smart contract development, such as using established libraries and conducting multiple rounds of auditing, can help mitigate these risks [13].

Encouraging widespread adoption of the blockchain-based property registry system is crucial for its success. This requires educating stakeholders about the benefits of the system and addressing any concerns they may have. User-friendly interfaces and seamless integration with existing systems can help in this regard. Additionally, demonstrating successful use cases and pilot projects can build confidence and drive adoption among real estate professionals, investors, and regulatory authorities [11].

By addressing these benefits and challenges, the blockchain-based property registry system can revolutionize the way properties are managed, enhancing security, transparency, efficiency, and inclusivity while overcoming the hurdles of scalability, regulatory compliance, and security [22].

## VIII. CONCLUSION AND FUTURE WORK

### A. Future Research and Development

Future work should address scalability challenges by exploring solutions like layer-2 protocols to manage high transaction volumes. A thorough analysis of regulatory compliance across various legal environments is needed to ensure broader applicability. Additionally, real-world implementation issues, including hardware limitations and security concerns, should be examined in detail. Simplifying technical explanations and including comparative analyses with existing methods could improve accessibility and contextualize the proposed system's benefits and trade-offs.

Subsequent investigations ought to concentrate on tackling the obstacles noted in this analysis, namely in relation to expandability and adherence to regulations. Investigating cutting-edge scalability options like Ethereum 2.0's sharding and layer-2 protocols (such as Plasma and state channels) can improve transaction throughput and lower latency [16]. Furthermore, creating regulatory compliance frameworks will be essential for broad adoption [15].

To increase accessibility and user experience, more work must be done on the User Interface (UI). Users will be able to manage their properties and tokens more effectively if the user interface is improved to facilitate smooth interaction with Ethereum wallets and offers more user-friendly navigation. In order to increase accessibility and convenience for users across a range of devices, future research should also investigate the integration of mobile applications.

Off-chain storage solution optimization is a significant topic for future effort. Leveraging decentralized storage technologies, such as the InterPlanetary File System (IPFS), can guarantee the efficient and secure storage of substantial metadata, such as photographs and property descriptions. In order to minimize the burden on the blockchain and preserve data accessibility and integrity, research should concentrate on enhancing the connection between off-chain and on-chain data. The robustness of the system can be further improved by investigating hybrid storage options and creating procedures for safe data retrieval and verification [21].

The property register system's capabilities will also be improved by looking into the possible integration of cutting-edge technology like artificial intelligence (AI) and the Internet of Things (IoT). Real-time data on property conditions can be obtained via IoT devices, allowing for automated registry updates and predictive maintenance. AI can help with property management decision-making, resource allocation optimization, and data analysis [14].

Future studies could also look into integrating more intricate DAO structures, as those made possible by Aragon-enabling platforms. Aragon offers customized governance structure decentralized organization creation and management tools. By introducing sophisticated features like multi-sig wallets, unique voting procedures, and role-based permissions, AragonDAO implementation may increase the property management system's complexity and level of control. As a result, more complex and adaptable governance models might be created to suit the different requirements and preferences of co-owners of real estate [9].

In conclusion, even if the suggested blockchain-based property register system has a number of advantages over existing ones, more investigation and development are required to resolve the issues raised and look into fresh avenues for improvement. The system may be made more resilient, effective, and future-ready by concentrating on scalability, regulatory compliance, UI enhancements, off-chain storage optimization, and the incorporation of cutting-edge DAO structures and upcoming technologies.

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